
Advancing Healthcare AI: Fine-Tuning Language Models for Accurate Medical Entity Extraction

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Abstract

This work focuses on developing a model to extract medical entities from unstructured free-text medical descriptions. The model is fine-tuned on a pre-trained LLM, combining the power of AI into this field. The model aims to act as a tool that could be utilized within business workflows to help automate and streamline data processing, decreasing human interference in this field. The work discusses the significance of the data and pre-trained model used, and proves its applicability in other fields. The model resulted in high BERT scores for evaluation. Ultimately, the model demonstrates high accuracy and flexibility, exhibiting the advantages of utilizing LLMs. This research was done to improve the fundamental process of technological development—data processing—hoping to contribute to societal benefit.

1 Introduction

The adoption of electronic medical records (EMR) and electronic health records (EHR), is a revolutionary transformation that greatly improved the efficiency and accuracy in diagnoses. EMR are like a hospital's digital record system for their own patients, while EHR gather a patient's health information across different providers [1]. The rate of implementing EMR systems in U.S. hospitals have risen from 6.6% to 81.2% between January and June 2020 [2]. Adopting digital records are crucial in the entire healthcare system for several reasons. It improves quality by reducing variability, standardizing, and streamlining data [3]. In terms of caring, it helps with more accurate data, greater decision-making, and up-to-date records [4]. In terms of administration, it facilitates access to records among hospital departments, billing and insurance companies, and patients [5].

While implementing electronic records can bring immense convenience to the industry and all general citizens, correctly and efficiently extracting medical data from free-text information has always been an overwhelming challenge among the industry [6]. In fact, according to research, physicians spend 49.2% of their total clinic time on EHR and desk work, compared to 27.0% spent on patients. Physicians also need to spend 1 to 2 after-work hours on EHR tasks [7]. Studies have shown that overload of administrative work, mainly authorization and documentation causes lower-quality care, lower career satisfaction, and higher possibilities of burnout [8]. Therefore, it is crucial to improve the efficiency of the process of extracting data into electronic records.

There are three general ways to extract medical entities from unstructured free-text medical descriptions. Manual data entry is the most traditional method. This method is time-consuming and costly [9]. It is also limited to the efficiency and the human error rate [10]. Another common method is utilizing traditional Natural Language Processing (NLP). For instance, Named Entity Recognition (NER), support vector machine (SVM), and Conditional Random Field (CRF) are popular tools used to achieve this goal [11] [12]. However, the approach also faces several challenges. For example, vocabulary and knowledge are limited to the corpus, which leads to difficulties in recognizing entities

that are not existing in the training data. To solve this problem requires a rich corpus, making the model harder to process [13]. Additionally, the model faces challenges in handling diverse or different styles, perspectives of expressions, such as formal and informal writing, technical and non-technical terminology, and different genres [14]. The other method is utilizing generative AI large language models (LLMs) to extract ideal entities. This work would discuss the performance of entity-extraction models fine-tuned from pre-trained LLMs.

2 Related Work

Several studies have successfully trained models to extract medical entities from free-text unstructured data [15][16]. Most of these tools utilize NLP techniques. A healthcare natural language API was also released by Google Cloud, demonstrating the potential of NLP in the medical field, but with limitations. For example, the inability to decide the extracted entities, the lack of adaptability to different context and terms, and its inability to extract phrases. These tools have improvement space. Utilizing AI large language models improves its flexibility and diminishes some of their limitations.

3 Methods

My model, which has similar objectives as earlier efforts, takes a different approach by fine-tuning LLMs for the task of medical entity extraction. This method enables my model to take advantage of the underlying LLM's tremendous language understanding skills, which have been trained on massive quantities of data, without requiring the same degree of training data for the specific entity extraction task. By building on the strong foundation of an LLM, my model can attain great performance while using fewer samples than training from scratch.

3.1 Generating training data

The success of AI models relies highly on the quality of training data, which can be challenging because of the access of resources, high costs, and privacy concerns. I intended to generate two versions of datasets: one consisting of common knowledge, and one for medical-specific purposes, which would fit with the model's initial purpose. Both datasets would be mainly generated by the OpenAI GPT-3 model, but would undergo data processing and cleaning afterwards. Manual reviewing and evaluating the rows, removing any duplicate, unsuitable, or irrelevant data were done at last. With limited resources, I built the first dataset of 1100 rows. The dataset includes common knowledge descriptions, questions, answers, and extracted entities organized in JSON format. After manual revision, the final dataset contained 1030 rows. The first dataset was created to verify the feasibility of the method, strategy, and pre-trained model. The strong results led me to decide against generating another dataset specifically for medical data, which would have required extensive data processing and manual checking. Instead, I generated a medical-specific dataset with only 50 rows for evaluation of the model considering the cost and time for the evaluation process.

The significance and value of the data could be discussed in further detail. Firstly, I will be discussing the decision of using JSON format instead of other data serialization formats. The advantages of JSON format demonstrated by various research studies [17][18], prove its standardized value and significance in entity extraction. The format's human readability, flexibility, platform independence, and efficient data storage make it an ideal choice for representing structured data in a compact and easily integrable manner, which is crucial for entity extraction tasks.

Secondly, The use of synthetic data in this project is highly valuable, as it provides a structured and machine-readable format for the information being processed. The recent research on synthetic data integration and its applications in various domains highlights the benefits of this approach [19][20].

- **Supplement Real-World Data:** Synthetic data can be used to supplement real-world data, providing a more comprehensive training dataset for AI models.
- **Enhance Data Security:** Synthetic data can enhance data security and privacy by reducing the need for sensitive information and minimizing privacy risks. It also allows the data to be public and shared with others.
- **Cost and Speed of Data Acquisition:** Synthetic data can be generated quickly and at a lower cost compared to collecting and annotating real-world data.

The use of synthetic data in AI training is significant because it can help address several challenges associated with real-world data collection. Recent research proving its feasibility opens a door for this training method. It offers a possible solution for generating datasets without direct access to sensitive information, simultaneously obtaining low cost yet high quality data.

3.2 Fine-tuning

The pre-trained models used were some of the most popular LLMs currently available, namely Mistral and Llama. The GPU used for training is NVIDIA_A100_80GB x 8, 640GB and approximately 3 hours in total. The model was computed on Google Colab.

My first fine-tuning approach involved fine-tuning the Mistral-7B-v0.1 base model using a combination of techniques, including:

- **PEFT (Parameter-Efficient Fine-Tuning):** This method enhances the model's performance and adaptability by fine-tuning a small subset of model parameters, significantly reducing computational requirements.
- **QLoRA (Quantized Low-Rank Adaptation):** This technique enables efficient fine-tuning of large language models by quantizing the model's weights and applying LoRA (Low-Rank Adaptation) updates.
- **Transformers Library:** To integrate these techniques, I utilized the Hugging Face library, which offers a seamless interface for fine-tuning pre-trained models. This allowed for efficient adaptation of the base model to specific task requirements.

However, during the fine-tuning process with the Mistral-7B-v0.1 base model, I encountered frequent errors, including hallucinations and garbled text. The training process took 13 minutes, with about 1 minute required to process each answer.

My second approach was done by fine-tuning the relatively new model, Mistral-7B-v0.2. I introduced optimization techniques INT4 quantization and mixed precision training (FP16) into the fine-tuning process. These techniques offer benefits such as reduced memory usage, accelerated training speed, and increased computational efficiency.

The model exceeded expectations. Firstly, during manual evaluation, which involved testing the model with 30 different prompts related to healthcare and common knowledge, previous errors such as hallucinations and garbled text did not occur. Secondly, the training time was notably reduced to 191 seconds, while the inference time remained around 1 minute.

For my third approach, I opted to fine-tune the Meta-Llama-3-8B base model. This decision was influenced by several factors: firstly, it was widely regarded as one of the most powerful large language models (LLMs) available at the time. Secondly, the parameter count of this model was closer to that of the Mistral base models, suggesting potential compatibility and comparability in performance. Additionally, there was a desire to minimize training time to conserve resources, prompting the selection of a model with a smaller parameter size. Finally, there was a contingency plan in place to try the larger 70-billion-parameter model if the fine-tuning attempt with Meta-Llama-3-8B proved unsuccessful.

The training time was only 147 seconds, with about 47 seconds to process every answer. The model obtained from fine-tuning the Meta-Llama-3-8B model was deemed acceptable, although they exhibited differences compared to those achieved through the fine-tuning of the Mistral-7B-v0.2 model. A thorough analysis and comparison of the outcomes from different fine-tuning approaches will be provided in the next section.

4 Results

With the first attempt (fine-tuning on the Mistral-7B-v0.1 base model) facing failures, the results would not be discussed. However, the results of fine-tuning the Mistral-7B-v0.2 base model and the Meta-Llama-3-8B base model is worth comparing.

Both models successfully extracted the expected entities in a correct format. The major difference is that the model fine-tuned on Meta-Llama-3-8B fails to give the answer the prompt asks for. It

directly gives the JSON data with one index containing the extracted entities within it whether I give a question and ask it to answer or not.

In contrast, the model fine-tuned on Mistral-7B-v0.2 more accurately follows the instruction: when asked a question the model replies with a extra index "answer"; when no question is given, it only returns the instructions and the entities extracted in JSON format. Therefore, this version among all of the fine-tuned models is preferred. An example of a general response by the preferred model could be shown on the README page on github containing the whole project.

The evaluation process utilizes BERT score for similarity comparison between the responses since the model gives open-ended responses. The evaluation results using the 50-rows medical description dataset had relatively high scores:

- **Precision:** 0.9795
- **Recall:** 0.9830
- **F1 score:** 0.9813

The biggest difference between the fine-tuned model and the baselines is the stability of its quality. The baseline model gives answers with different kinds of formats: it sometimes gives answers, sometimes responses all the entities as columns, sometimes adds a column "return": "Json". Additionally, most baseline models tend to produce verbose answers with unnecessary wording in the responses. In contrast, the fine-tuned model consistently provides standardized, concise responses that are easier to manipulate.

5 Discussion

5.1 Limitations

Since the data used in this model were synthetic and contained no real-life data, this introduces limitations on the potential improvements achievable if trained on real-world data. If real-life private data from organizations or hospitals were organized into training datasets and used to fine-tune LLMs using this method, it could create models that are tailor-made for specific needs, enhancing both accuracy and privacy.

Furthermore, attempts to fine-tune on other pre-trained LLMs might possibly improve the quality of the models; however, it requires several additional experimenting and adjusting.

5.2 Broader impacts

The model and process method prove its potentiality in different areas, including legal, financial, and administrative sectors. The model serves as a useful tool that can be integrated with PDF readers, speech-to-text technologies, and other applications.

The concerns of the model is its might its influence in industries if it responses in misleading or inaccurate content. However, based on the results and testings

In the legal sector, the model could help with contract analysis, case law summaries, and regulatory compliance. with the financial sector, it could aid with the extraction of indicators from earnings reports and investment assessments. Administratively, it could efficiently handle data from HR records and government forms.

The model's integration with PDF readers and speech-to-text techniques expands its applicability in these areas. PDF reader plugins allow organized data and key points extracted from documents as they are viewed. When combined with speech-to-text technology, the model allows for real-time transcription and extraction of crucial information from audio, which is invaluable for meetings and court depositions.

This methodology improves efficiency by automating data extraction and reducing human error with high accuracy. Its combination with existing technologies and workflows increases its adoption potential, making it a useful tool for businesses hoping to automate and streamline document processing. Overall, the model's versatility and efficiency make it revolutionary, freeing up human resources and maximizing their value.

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